Retail Apparel Performance Review

Associate Data Analyst Assessment - Cargills

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Objective & Approach

What we analyzed, and how

6

months of data

Jan–Jun 2025, across sales, foot traffic, inventory & promos

9

stores

Spanning 4 regions, with two stores opening/closing mid-period

10

business questions

Answered in Python, backed by KPIs and statistical tests

Approach: Clean and validate the raw data in Python -> calculate KPIs for each business question -> build a single Power BI dashboard with tailored views for Store Managers, the VP of Merchandising, and Senior Operations.

Data Preprocessing

Cleaning steps that changed the analysis

Removed 347 exact-duplicate rows

20,774 -> 20,427 transaction lines after dedup.

Caught a date-format contradiction

The brief specifies DD/MM/YYYY, but validation showed the first component never exceeds 6 (matching the Jan-Jun window) while the second exceeds 12 in 2,655 rows - the real format is MM/DD/YYYY. Flagged for the assessor.

Fixed a duplicate SKU name

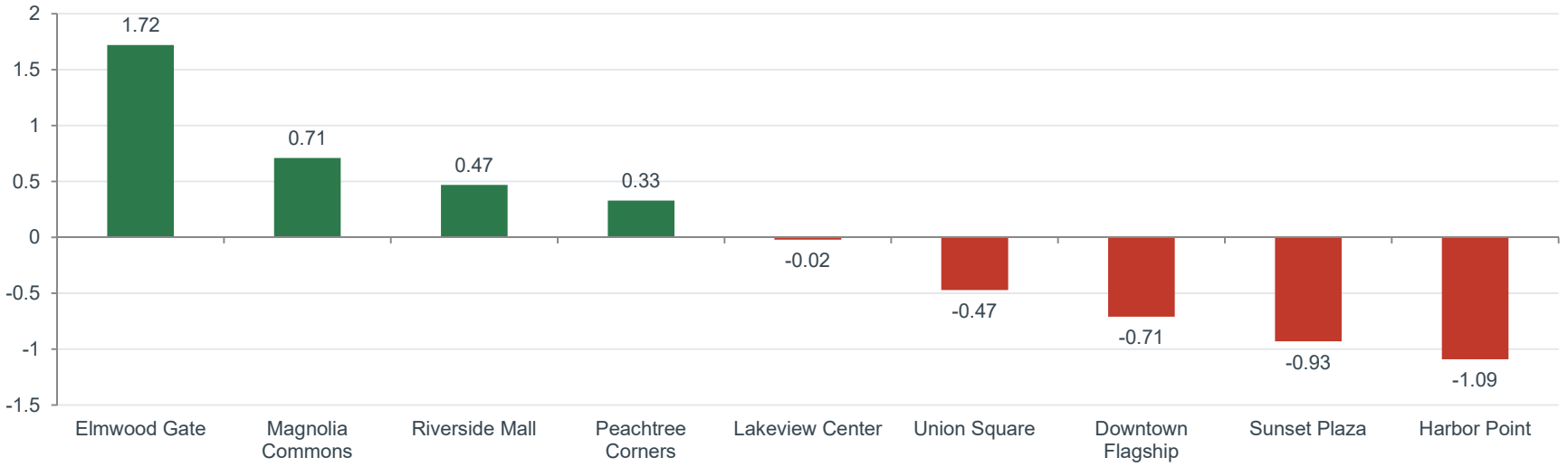
Two different product codes (P001, P010) both read “Rain Jacket” - renamed distinctly before aggregating.

Imputed missing values transparently

437 missing discount %, 9 missing customer IDs - each imputed row is flagged, not silently guessed.

Overall Store Performance

A composite score keeps store size from skewing the ranking



Score blends revenue/sqft/day, conversion rate, and margin (z-scored) so a small store beating expectations outranks a large one merely coasting on footprint.

Real Decline vs. One-Off Dip

Not every bad month is a problem - and not every new store is fine

Harbor Point

Genuine declining trend

Consistent negative slope across all 6 months (-7.5% of its average per month). This is a structural problem, not noise - needs intervention.

Elmwood Gate

Watch, don't act yet

Opened in April with only 3 months on record. Its sharp early dip is more likely post-launch normalization than a lasting decline - revisit after another quarter of data.

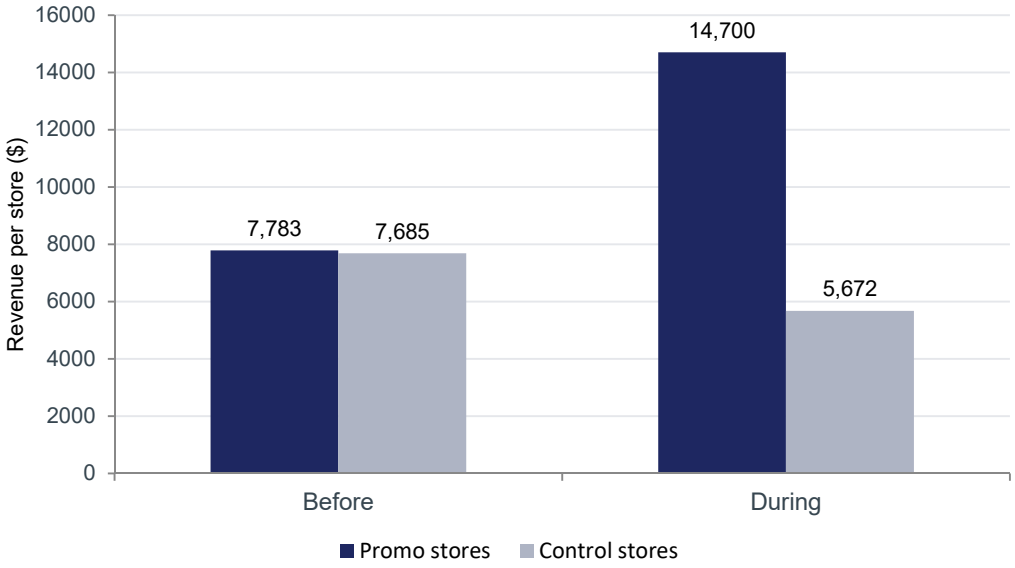
The other 7 stores

One-off bad month

Trend lines are stable; any single weak month sits within normal month-to-month noise, not a sustained slide.

Spring Outerwear Promotion - Repeat It

Diff-in-difference against non-promo stores isolates the true lift



+88.9%

Promo stores, before -> during

-26.2%

Control stores, seasonal baseline

t = 5.54, p < 0.0001

Uplift is statistically significant

+$14,585

Margin gain, promo stores

Recommendation: repeat the promotion - the lift is statistically significant and margin-positive, not just a revenue shift from discounting.

SKU & Pricing Actions

Where the margin is thin and discounting is doing the heavy lifting

Canvas Sneakers (Footwear)

Recommendation: Reprice

96.6% of its revenue only sells on discount, at **34.7%** margin - the lowest of any SKU on the floor. It isn't earning its shelf space at full price.

Footwear needs a category-wide look

Footwear has the widest margin spread of any category (34.7% - 57.4%), versus a tight 1.6-point spread in Bottoms.

That inconsistency - not any single SKU - is the real repricing opportunity.

Limitation: true competitive pricing (Q5) needs external market-price data we don't have. What's shown here is an internal margin-outlier proxy, not a market comparison.

Loyalty Pays Off; Conversion Is Climbing

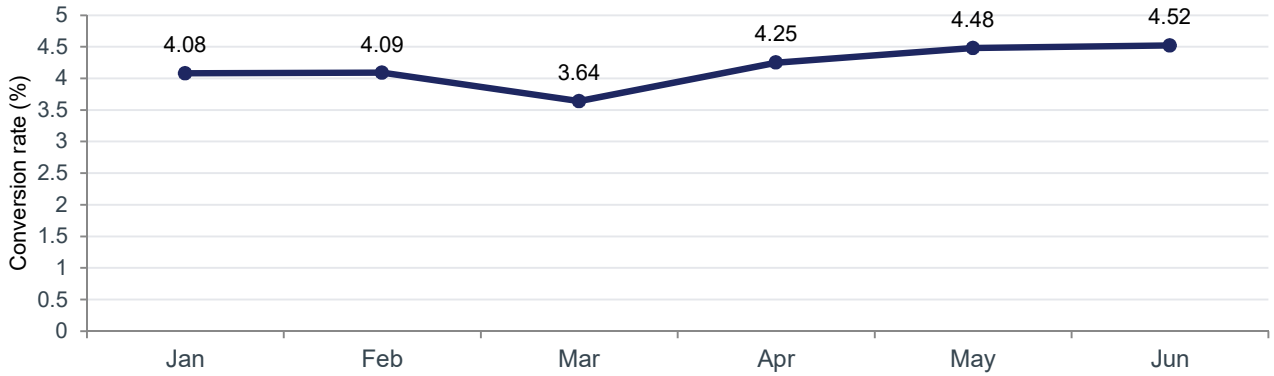
$159 vs $123

Avg. order value: loyalty vs. non-loyalty (+29%)

96.9% vs 56.0%

Repeat purchase rate: loyalty vs. non-loyalty

Both differences are statistically significant (t = 18.08, p < 0.0001) - loyalty members aren't just self-selected big spenders by chance.



Inventory Imbalances & Regional Allocation

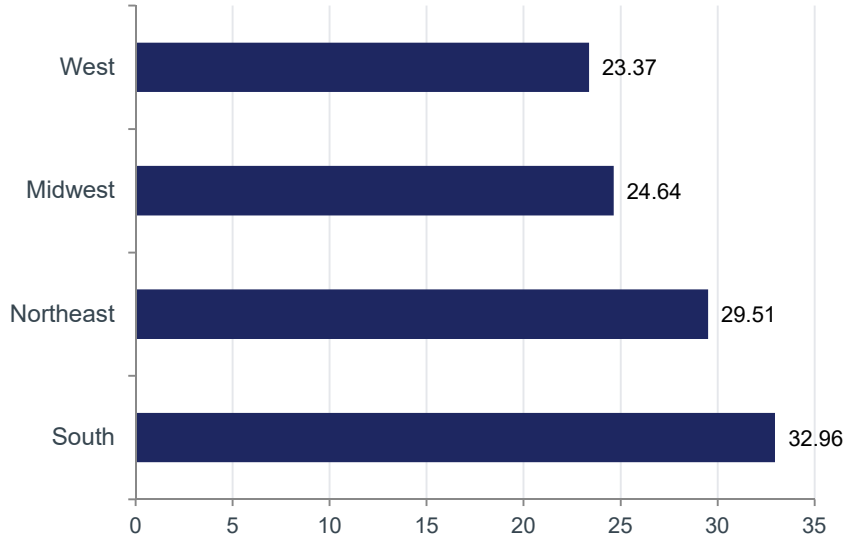
Overstocked: Elmwood Gate (new store)

Nearly every SKU there carries 5.7-7.0 months of supply - stock levels were set before real demand data existed and haven't been adjusted since launch.

Understocked: fast movers at ST06-ST08

Polo Shirt, Slim Jeans, and Flannel Shirt are selling through in under 2.2 months of supply at Harbor Point, Magnolia Commons, and Peachtree Corners - restock risk if not replenished soon.

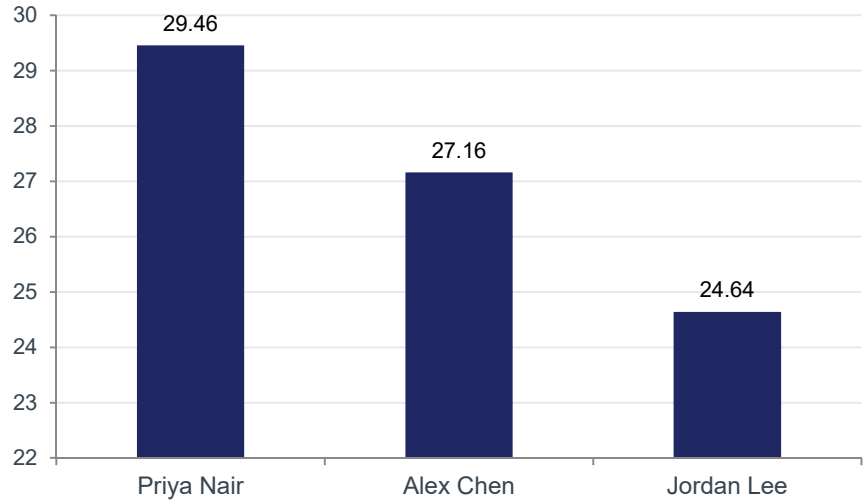
Revenue per sq ft by region



West carries the most square footage per dollar earned - the clearest candidate for space rebalancing.

Manager Performance & Recommendations

Revenue per sq ft by manager (3 stores each)



Margin runs ~56% for all three managers, so the gap here is operating efficiency, not merchandise mix.

Next steps

1. Repeat the Spring Outerwear promotion; consider extending it to adjacent categories next season.
2. Reprice Canvas Sneakers and audit other high discount-dependency SKUs in Footwear.
3. Right-size Elmwood Gate's inventory downward as real demand data accumulates; prioritize restock at Harbor Point, Magnolia Commons, and Peachtree Corners.